ECON 0100 | Vignette E2 | Competition

Solutions

Answers and work are shown in the red blocks.

Q1 | A Higher Marginal Cost

The Witching Hour has a marginal cost of $MC = 10 + 2q$. Many identical workshops sell identical cauldrons, so each takes the market price as given.

a) At $P = 50$ what quantity maximizes the firm's profit?

Profit is largest where $P = MC$: $50 = 10 + 2q$, so $q = 20$.

b) Calculate the firm's maximal profit if $ATC = 50$ at the profit-maximizing quantity.

$(P - ATC) \times q = (50 - 50) \times 20 = 0$. Zero economic profit: the market is in long-run equilibrium.

Now suppose the Ministry taxes every cauldron 10 coins, so marginal cost rises: $MC = 20 + 2q$. Start with identical firms earning zero economic profit in long-run equilibrium.

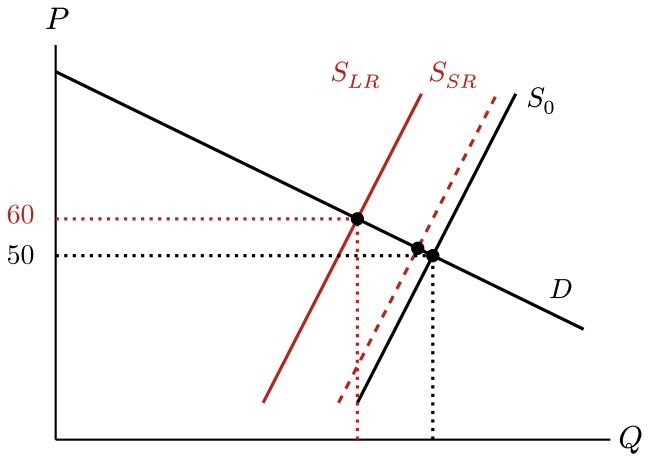
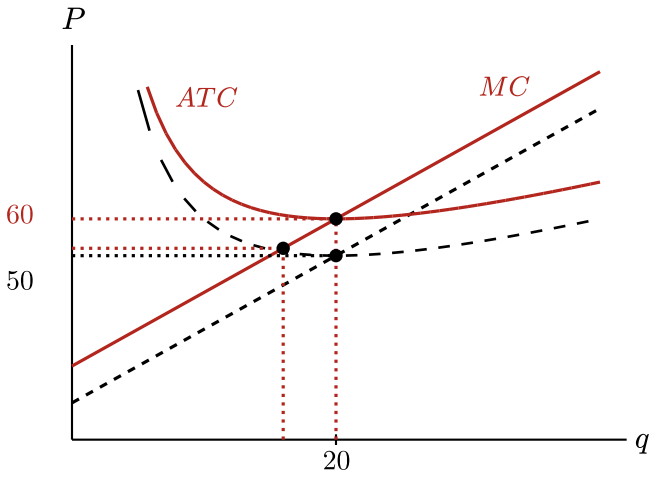
c) What happens to price, each firm's output, and profit after the new cost?

Every workshop's MC and ATC shift up by 10, so market supply shifts up. The price rises, but by less than 10 while the same workshops are all still selling. At the new price each workshop makes fewer cauldrons, and the price is below its new ATC, so it makes a loss.

d) In the long run the price rises to 60. What quantity maximizes each remaining firm's profit?

$P = MC$ with the new marginal cost: $60 = 20 + 2q$, so $q = 20$, the same quantity as before the tax.

e) What happens in the long run to price, each remaining firm's output, total market output, and the number of firms?

Losses make workshops leave. Market supply shifts in and the price rises until the remaining workshops earn zero economic profit again, at $P = 60$: the price has risen by the full 10. Each remaining workshop is back at its original output of 20, total market output is lower because

buyers want less at 60 than at 50, and there are fewer workshops. (Left graph: the market. Right graph: one firm.)